

YEAR 7 • UNIT 3 • EXTRA HOMEWORK

The Point Never Moves

Reading decimals • multiplying and dividing by 10 • rounding • mg, g, kg and t

Name: _____

Class: _____

Date given: _____

Date due: **Monday 28 September 2026****Why this sheet exists**

0.46, 4.6, 46 and **460** are the **same two digits**. Nothing about the 4 or the 6 is different. What tells you which number you are looking at is **where the point is**.

So when somebody says “multiply by 100”, the point **stays exactly where it is** and the **digits move** two columns to the left. That is why 7.2×10^3 is 7200 and not 7.2000 – and you can watch it happen in the tables on this sheet.

No calculator. Write in every box.

Part 1 • Where the point is

X1 One digit per column

Put each number into the table. One digit goes in each column. The first one is done for you.

hundreds	tens	ones	.	tenths	hundredths	thousandths
47.3						
	4	7	.	3		
6.85						
			.			
130.06						
			.			
0.904						
			.			

X2 The same digit, five different jobs

Every number below has a **6** in it, and the 6 is worth something different every time. Join each number to what its 6 is worth. The first one is done for you.

Word help – the columns after the point

tenths the **first** column after the point. 0.6 is six tenths

hundredths the **second** column. 0.06 is six hundredths

thousandths the **third** column. 0.006 is six thousandths

Number		What the 6 is worth
1. 6.2	C	A six tenths
2. 0.63	☐	B six hundredths
3. 0.86	☐	C six ones
4. 0.006	☐	D six tens
5. 62.5	☐	E six thousandths

X3 Saying it out loud**After the point, one digit at a time**

12.34 is “**twelve point three four**”. **Not** “**twelve point thirty-four**”.

The digits after the point are read **one by one**, so a zero has to be said: 4.05 is “**four point zero five**”.

Number	Say it
12.34	twelve point three four
5.9	
0.71	
8.06	
	three point one four
	twenty point zero five

X4 Which is bigger?

Write < or > in each box.

A longer number is not a bigger number

0.65 has more digits than 0.7, and it is **smaller**. Compare the **tenths first**: 6 tenths against 7 tenths. So $0.65 < 0.7$.

(a) $0.4 \square 0.35$

(c) $2.5 \square 2.45$

(e) $7.30 \square 7.3$

(b) $0.8 \square 0.81$

(d) $0.09 \square 0.1$

(f) $0.62 \square 0.6$

One pair in that list is **neither** bigger nor smaller. Which one, and why?

Part 2 • Moving the digits
X5 Watch it happen

Each row is the row above it multiplied by **10**. Every digit moves **one column to the left**. Fill in the rest of the table.

hundreds	tens	ones	.	tenths	hundredths	thousandths
		4	.	6		
	4	6	.			
			.			

(a) Write the three numbers. 4.6, _____, _____

(b) The 4 and the 6 are in all three rows. What moved, and what stayed still?

X6 Multiplying moves the digits left

One place for 10, two for 100, three for 1000.

(a) $3.2 \times 10 = \underline{\hspace{2cm}}$

(d) $3.2 \times 100 = \underline{\hspace{2cm}}$

(g) $3.2 \times 1000 = \underline{\hspace{2cm}}$

(b) $0.7 \times 10 = \underline{\hspace{2cm}}$

(e) $0.7 \times 100 = \underline{\hspace{2cm}}$

(h) $0.05 \times 100 = \underline{\hspace{2cm}}$

(c) $5.48 \times 10 = \underline{\hspace{2cm}}$

(f) $6.9 \times 100 = \underline{\hspace{2cm}}$

(i) $1.28 \times 1000 = \underline{\hspace{2cm}}$

X7 Dividing moves the digits right

The zeros you have to write yourself

$7 \div 100$. The 7 moves two columns right, past the ones and past the tenths, and lands in the **hundredths**. Nothing is left in the ones or the tenths, so you write **0** in both:
 $7 \div 100 = 0.07$.

- (a) $46 \div 10 =$ _____ (d) $46 \div 100 =$ _____ (g) $46 \div 1000 =$ _____
 (b) $8 \div 10 =$ _____ (e) $8 \div 100 =$ _____ (h) $9 \div 1000 =$ _____
 (c) $520 \div 10 =$ _____ (f) $520 \div 100 =$ _____ (i) $3.5 \div 100 =$ _____

X8 The little raised number**The power counts the zeros**

10^2 means $10 \times 10 = 100$. Two zeros. 10^3 means $10 \times 10 \times 10 = 1000$. Three zeros.
 Say it aloud: 10^2 is “**ten squared**”, 10^3 is “**ten cubed**”, 10^4 is “**ten to the power of four**”.

Power	Written out	Say it
10^2	100	ten squared
10^3		
10^5		
	10 000	

X9 Now with the power written in

Same job as X6 and X7. The power tells you **how many places**.

The folk rule is wrong

“To multiply by 10, add a zero” works for whole numbers and **nothing else**. 7.2×10^3 is **7200**. **Not 7.2000, which is just 7.2.**

- (a) $2.4 \times 10^2 =$ _____ (d) $830 \div 10^2 =$ _____
 (b) $0.06 \times 10^3 =$ _____ (e) $6 \div 10^3 =$ _____
 (c) $15 \times 10^4 =$ _____ (f) $2.5 \div 10^1 =$ _____

X10 Which way, and how far?

Look at what happened to the digits. Write **left** or **right**, and how many places. The first one is done for you.

From		To	Left or right?	How many places?	So you
4.8	→	480	left	2	$\times 10^2$
6.1	→	0.061			
90	→	9000			
7	→	0.7			

Part 3 · Stopping somewhere**X11 Which end is it nearer?**

The line below runs from 3 to 4, and it is marked in **tenths**.



(a) Write the number that **P** is marking. _____

(b) Write the number that **Q** is marking. _____

(c) Is **P** nearer to 3 or to 4? _____

(d) Is **Q** nearer to 3 or to 4? _____

X12 Round to the nearest whole number

Look at the next digit only

Look at the **tenths** digit. 0, 1, 2, 3, 4 – round **down**. 5, 6, 7, 8, 9 – round **up**.

12.3 → **12** 12.8 → **13** 12.5 → **13**

(a) 6.2 → _____

(c) 14.5 → _____

(e) 29.8 → _____

(b) 8.9 → _____

(d) 0.4 → _____

(f) 0.61 → _____

X13 Counting decimal places

Decimal places are the digits **after the point**. Start counting **at the point**, not at the front of the number.

Number	How many d.p.?	Number	How many d.p.?
5.28	2	130.4	
0.9		7.0625	
12.346		86	

X14 Round to 1 decimal place

Keep **one** digit after the point. Look at the **second** digit after the point to decide.

- (a) 4.62 → _____ (c) 0.348 → _____ (e) 9.44 → _____
 (b) 8.17 → _____ (d) 23.75 → _____ (f) 0.251 → _____

X15 The zero that has to stay

35 and 35.0 are not the same answer

34.98 to **1 decimal place**. The 8 rounds the 9 up, the 9 carries into the 4, and you get **35.0**.

Write **35.0**, not 35. The **.0** is what shows you rounded to **one** place — and it is the whole mark.

Round each one to **1 decimal place**. Every answer on this line ends in a zero.

- (a) 6.97 → _____ (c) 19.98 → _____
 (b) 0.96 → _____ (d) 2.04 → _____

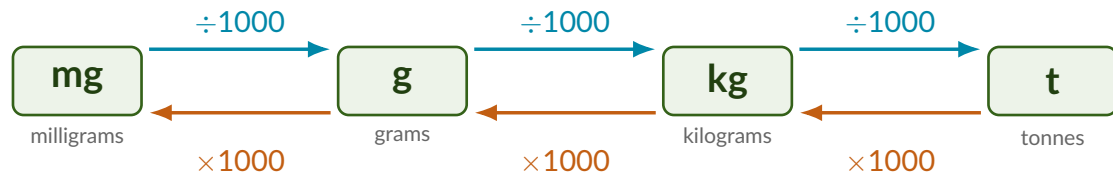
X16 Round to 2 decimal places

Keep **two** digits after the point. Look at the **third**.

- (a) 3.472 → _____ (c) 12.005 → _____
 (b) 0.1849 → _____ (d) 8.996 → _____

Part 4 • The mass ladder

There are four units of mass and they are **three identical steps** apart. Every step is $\times 1000$ one way and $\div 1000$ the other – which is the digits moving **three places**, exactly as in Part 2.



X17 One step

Use the ladder. Each of these is **one** step.

- | | |
|-----------------------|----------------------|
| (a) 5 kg = _____ g | (d) 750 g = _____ kg |
| (b) 2000 mg = _____ g | (e) 1.8 g = _____ mg |
| (c) 0.4 t = _____ kg | (f) 60 kg = _____ t |

X18 Two steps

Two steps is $\times 1000$ twice

kg to mg goes **kg** $\rightarrow$ **g** $\rightarrow$ **mg**: $\times 1000$, then $\times 1000$ again. That is $\times 10^6$ – the digits move **six** places.

$$4 \text{ kg} = 4000 \text{ g} = 4\,000\,000 \text{ mg}$$

- | | |
|---------------------|---------------------------|
| (a) 3 kg = _____ mg | (c) 500 000 mg = _____ kg |
| (b) 2 t = _____ g | (d) 0.25 t = _____ g |

X19 Which is heavier?

You cannot compare two numbers until they are in the **same unit**. Change one of them, then write $<$ or $>$.

- (a) 400 g or 0.5 kg ?

same unit: _____

☐

- (b) 2500 mg or 2 g ?

same unit: _____

☐

- (c) 0.9 t or 950 kg ?

same unit: _____

☐

X20 Mr Bowen's homework

Only **one** of these six is right. Find it, and correct the other five.

- (a) $4.6 \times 100 = 4.600$ _____ (d) 130.4 has 4 d.p. _____
- (b) $7 \div 100 = 0.7$ _____ (e) $3 \text{ kg} = 3000 \text{ mg}$ _____
- (c) 0.96 to 1 d.p. = 1.0 _____ (f) $0.65 > 0.7$ _____

X21 The whole sheet in one question

Mr Bowen's medicine bottle says **0.045 kg**.

- (a) Write that mass in **grams**. _____
- (b) Write it in **milligrams**. _____
- (c) Round your **gram** answer to **1 decimal place**. _____

Show the steps of the ladder here.

(d) The digits **4** and **5** are in all three of your answers, and nothing else about them is the same. Say in one or two sentences what changed.